

LangGraph Report

Metrics

```
[
{
  "name": "Accuracy",
  "why_best": "Provides an overall measure of how often predictions match labels; suitable as the first metric when dealing with balanced datasets.",
  "whentouse": "Use for classification tasks where classes are well-balanced and you need a quick, general idea about model performance."
},
{
  "name": "Precision",
  "why_best": "Measures exactness or how many selected items from all positive predictions is relevant; important when false positives have high costs.",
  "whentouse": "Use in scenarios where the cost of misclassification as a negative (false negatives) outweighs that of missing actual cases."
},
{
  "name": "Recall",
  "why_best": "Measures completeness or how many correct positive results are identified; crucial when false negatives have high costs.",
  "whentouse": "Use for situations where the cost of misclassification as a true negative (false positives) outweighs that of missing actual cases."
},
{
  "name": "F1 Score",
  "why_best": "Harmonizes precision and recall into one metric; best when seeking balance between false positives and negatives in imbalanced datasets.",
  "whentouse": "Use for unbalanced classification problems where both False Positives (Type I Errors) and False Negatives (Type II errors) are of concern."
},
{
  "name": "Mean Squared Error",
  "why_best": "Measures the average squared difference between predicted values by a model or estimator and actual values; best for regression tasks.",
  "whentouse": "Use as an evaluation metric in supervised learning problems where you need to quantify prediction errors."
}
]
```

Explanation

Sure! Let's break down and explain the provided metrics with detailed explanations, examples, and key considerations.

Metrics:

Accuracy

Explanation:

Accuracy measures how many predictions were correct out of all made by a model in classification tasks.

$$\text{Accuracy} = (TP + TN) / (TP + FP + FN + TN)$$

where TP is True Positives, TN is True Negatives; FP and FN are False Positives and False Negatives respectively.

Example:

Consider an email spam filter that correctly identifies 80% of actual emails as not being spam.

- If it predicts positively (not a real spam): Correctly identified non-spam = TP + TN
- Incorrect predictions made to either type:
- Non-spam predicted incorrectly (+1) -> False Positive (FP)
- Spam wrongly marked as non-spam (-1) -> False Negative (FN)

Key Considerations / Pitfalls:

Accuracy can be misleading in imbalanced datasets where there are many more instances of one class. High accuracy might still occur if the model is biased towards predicting only a single type.

Precision

Explanation:

Precision measures how accurate positive predictions were.

$$\text{Precision} = TP / (TP + FP)$$

Example:

Imagine an ad targeting system that sends ads to users it thinks will buy. If 80% of targeted individuals are actually buyers, but only half correctly identified as potential buyers:

- True Positives (correctly predicted interested): 40
- False Positives (incorrect positives): 20

$$\text{Precision} = TP / (TP + FP) = 40 / (40 + 20) = 0.67$$

Key Considerations / Pitfalls:

High precision does not necessarily mean high accuracy, especially if the dataset is imbalanced.

Recall

Explanation:

Recall measures how well a model can identify all relevant instances.

$$\text{Recall} = TP / (TP + FN)$$

Example:

Continuing with our ad targeting system:

- True Positives identified correctly as interested buyers remain 40
- False Negatives missed identifying actual potential buyer group of size x:

If there are only y people who actually bought the product, but we predicted they would not be an interest (FN):

$$\text{Recall} = TP / (TP + FN) = 40 / (40 + z)$$

Key Considerations / Pitfalls:

High recall does imply that many actual positives were captured. However, it also means higher false negatives which can affect performance in scenarios where missing true cases is costly.

F1 Score

Explanation:

F1 score harmonizes precision and recall into a single metric.

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

Example:

Using our previous example's values,

- Precision was found as 0.67, while
- If we had an additional FN count such that y=60 and x becomes z thus impacting recall calculation.

$$F1 = 2 \frac{0.67 \times Recall}{0.67 + Recall}$$

Key Considerations / Pitfalls:

Balancing precision with recall is crucial for achieving a fair F1 score, especially in imbalanced datasets where one metric might be significantly higher than the other.

Mean Squared Error**Explanation:**

Mean squared error measures how close predictions are to actual values by averaging squared differences between predicted and true outcomes.

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

where $\hat{y}$ is prediction, n the total number of observations.

Example:

In a weather forecasting model predicting temperatures:

- Actual Temperature [20°C] vs Predicted Temperatures [22°C]

$$MSE = \frac{1}{n} \sum ((20 - 22)^2 + \dots)$$

Assuming predictions were accurate for other days too.

If the sum of squared differences across all n observations is x, then:

$$MSE = x / n$$

Key Considerations / Pitfalls:

MSE can be sensitive to outliers; extremely high errors will disproportionately affect it. It's also essential as an evaluation metric in regression tasks.

In summary:

- **Accuracy:** Quick measure but not ideal for imbalanced datasets.
- **Precision & Recall:** Evaluate different aspects of performance, with trade-offs between them being considered through F1 score when needed.
- **F1 Score:** Balances precision and recall; best used where both types are important in unbalanced scenarios.

Mean Squared Error (MSE) is specific to regression tasks for measuring prediction errors. Understanding these metrics helps choose the appropriate one based on dataset characteristics, business context of predictions or outcomes being measured accurately reflecting model performance.

Time used: 118.36 seconds

Token usage: LangGraph + LangSmith Workflow

Token Usage Details

Per LLM Call

Step	Input Tokens	Output Tokens	Total Tokens
Selector	82	354	436
Explainer	410	997	1407
Formatter	1435	1003	2438

Totals

Prompt Tokens	Completion Tokens	Total Tokens
1927	2354	4281

Performance

Elapsed (s)	Completion tok/s	Total tok/s
118.14	19.92	36.24